

13. Part 1:

Annotation & Syntax Reference

1. Spring Annotation Cheatsheet

List all of the **annotations** you've learned from **class** and **homework** in a file named **annotations.md**. This will serve as your **Spring annotation reference**.

Part 2:

Hands-on Project Work

2. Setup Starter Project & Test APIs

- Clone the following **repository**:
https://github.com/CTYue/springboot-redbook/tree/05_02_slides JPQL_EntityManager_Session
- Make the necessary **code/configuration changes** to bring up the **application**.
- Test each **controller** using **Postman** and take **screenshots**.

3. Write Custom JPA Methods and Validate Syntax

- Based on the codebase from **Task 2**:
 - Write custom **JPA methods** using **Spring Data JPA naming conventions**.
 - Demonstrate how **JPA** performs **compile-time syntax checks** on method names.

4. Implement JPQL and Native SQL Queries

- In your **repository interface**, add:
 - One **JPQL query** using the **@Query** annotation
 - One **native SQL query** using **@Query(nativeQuery = true)**
 - Update your **service layer** to use these queries.
 - Test both using **Postman** and capture **screenshots**.
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Part 3:

Core Conceptual Understanding

5. Technology Comparison

Explain the differences and relationships among:

- **Spring Data JPA**
- **Hibernate**
- **HikariCP**
- **JDBC**

6. JPA Relationships

Explain the following annotations with examples:

- **@OneToMany**
- **@ManyToOne**
- **@ManyToMany**

Also describe and explain the **configuration properties** inside each, such as **mappedBy**, **cascade**, and **fetch**.

7. Cascade Options in JPA

Explain **cascade = CascadeType.ALL** and **orphanRemoval = true**.

Also describe all other **CascadeType** values:

- **PERSIST**
- **MERGE**
- **REMOVE**
- **DETACH**
- **REFRESH**

8. Fetch Types in JPA

Explain **FetchType.LAZY** and **FetchType.EAGER**, and describe **when to use each**.

Part 4:

Querying and Data Access

9. JPQL Overview

Explain what **JPQL** is and how it differs from **SQL**. Include **examples**.

10. Using the @Query Annotation

Explain what the **@Query** annotation does, where it is used, and how to use it for both **JPQL** and **native queries**.

Part 5:

Hibernate and Entity Management

11. EntityManager vs SessionFactory

Compare **EntityManager** and **SessionFactory**.

Include **screenshots** from the **codebase** to illustrate their relationship.

12. SessionFactory vs Session

Explain the role of **Session** and how it differs from **SessionFactory**.

Part 6:

Database Fundamentals

13. SQL Transactions

Explain what a **transaction** is in the context of **relational databases**, and how **Spring/Hibernate** manage **transactions**.

14. Hibernate Caching

Explain **Hibernate's caching mechanisms**.

Compare:

- **First-Level Cache** (session scope)
- **Second-Level Cache** (shared/global scope)